

Ethan Waxman

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EDUCATION

Cornell University, College of Engineering — B.S. Computer Science & Applied Physics, GPA: 3.53
Honors: 4-Year Varsity Heavyweight Rower (balanced 20+ hrs/week training with rigorous double major)
Certifications: AWS Solutions Architect — Associate (Jun. 2026)
Relevant Courses: Foundations of AI, Computational Linguistics, Neural Networks, Compilers, Computer Architecture, Operating Systems, Discrete Structures (Honors), Functional Programming

ML & DATA ENGINEERING EXPERIENCE

- FI Consulting, Arlington, VA** Software Developer Jun. 2024 – Present
- Led design of 3 Graph Neural Network (GNN) architectures for recommendation and churn prediction; achieved 82% prediction accuracy identifying \$100M+ in at-risk revenue; collaborated with clients to iterate and orient the project.
 - Architected and optimized GNN training pipelines using PyTorch to process 1M+ nodes/users in under 5 minutes, overcoming high-dimensional data scaling bottlenecks.
 - Owned an ML entity resolution service on AWS SageMaker from development to deployment, utilizing fuzzy matching and probabilistic record linkage to remove 10,000+ duplicates.
 - Engineered high-throughput ETL pipelines using Apache Airflow to migrate multiple terabytes of data into MongoDB for downstream model consumption.
 - Developed an executive analytics dashboard using PySpark and Databricks, enabling real-time distributed querying of 15M+ customer records.
 - Deployed a Jenkins pipeline to automate CI/CD; dropped build and deployment times by 75%.
- Machine Learning Intern Jun. 2023 – Jun. 2024
- Reduced inference latency for a RAG-based search application by 45%, implementing Redis caching to ensure high reliability and sub-second response times for 50+ concurrent users.

RESEARCH AND ML PROJECTS

- Large Language Model (LLM) Cognition Research** — Cornell University
- Designed an experimental framework to evaluate LLM interpretability using controlled syntactic ambiguity tasks; analyzed correlation between next-token probabilities and human repair preferences.
 - Authored a 7-page technical paper analyzing how model architecture affects the processing of syntactic ambiguity.
- Iterative Chess Engine** — Python, PyTorch, Reinforcement Learning
- Developed 4 engine iterations, evolving from Minimax with Alpha-Beta pruning and MCTS to a hybrid deep learning evaluation model.
 - Trained a Policy-Value Neural Network in PyTorch using 100,000 Grandmaster positions (100 epochs) and reinforced through thousands of self-play iterations.
- Full-Stack Academic Compiler** — Java
- Led a 3-person team to engineer a 10,000+ line compiler supporting 15+ language features including recursion, arrays, and lexical scoping.

TECHNICAL SKILLS

ML Frameworks: PyTorch, TensorFlow, Keras, Scikit-learn, NLTK

Data Tools: Apache Airflow, PySpark, Databricks, MongoDB, Redis, ChromaDB

Cloud & DevOps: AWS (Solutions Architect – Associate, SageMaker, Certified Cloud Practitioner), Docker, Neo4j, Terraform, Jenkins

Languages: Python (Advanced), Java (Advanced), OCaml (Advanced), Rust (Intermediate), C++ (Intermediate)